

Program: Diploma in Communication and Computer Networking	
Course Code: 3321	Course Title: Computer Architecture and Organization
Semester: 3	Credits: 4
Course Category: Programme core course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Identify the basic structure and functional units of a digital computer.
- Understand Memory Management, External Memory, I/O Devices etc.
- Understand Processor Structure, Functions and Register Organization.
- Understand Control Unit Organization, Hardwired and Micro-programmed control units.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge in Computer systems		Introduction to IT systems Lab	1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Understand Von Neumann Machine.	15	Understanding
CO2	Acquire knowledge about basic memory organization.	14	Apply
CO3	Develop knowledge about processor structure and functions.	16	Apply
CO4	Implement the Control Unit.	13	Apply
	Series Test	2	

CO–PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1				
CO2	3	1	2				
CO3	3	2	1				
CO4	3	2	1				

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Understand Von Neumann Machine.		
M1.01	To describe Von Neumann Machine	3	Understanding
M1.02	To explain various Computer functions	3	Understanding
M1.03	To describe Interconnection structures	3	Understanding
M1.04	To describe Bus Interconnection	4	Applying
Contents: Basic Computer Architecture The Von Neumann Machine – Computer Components - Computer functions – Instruction Fetch and Execute – Interrupts – I/O Function- Interconnection structures – Bus Interconnection Structures.			
CO2	Acquire knowledge about basic memory organization.		
M2.01	To describe the organization of Magnetic Disk	7	Understanding
M2.02	To explain Optical Memory	7	Understanding
	Series Test –I	1	

Contents: Basic Memory Organization Characteristics of Memory System –The Memory Hierarchy - Cache Memory Principles -Types of ROM & RAM. Magnetic Disk– Data Organization and formatting – Optical Memory – Compact Disk – Digital Versatile Disk– High Definition Optical Disks.			
CO3	Develop knowledge about processor structure and functions.		
M3.01	To describe Processor organization.	3	Understanding
M3.02	To illustrate Register organization	4	Understanding
M3.03	To explain Instruction Cycle	5	Applying
M3.04	To explain Instruction Pipelining	4	Applying
Contents: Processor Structure Processor organization - Register organization – Control and Status Registers- Instruction Cycle – Instruction Pipelining.			
CO4	Implement the Control Unit.		
M4.01	To describe Micro programmed control	7	Understanding
M4.02	To describe Hardwired Control	6	Applying
	Series Test–II	1	
Contents: Control Unit Organization Micro operations – Fetch Cycle – Indirect Cycle - Interrupt Cycle – Execute Cycle – Instruction Cycle - Control of the Processor - Hardwired implementation - Micro programmed control.			

Text/Reference

T/R	Book Title / Author
T1	Computer Organization - Carl Hamacher- Mc Graw Hill, fifth edition.
T2	Digital fundamentals – Thomas Floyd & R.P. Jain, Pearson Education(2005)

T3	Digital Design, M. Morris Mano & Michael D. Ciltti, Pearson Education, 5 th Edition
R1	Computer Organization and Architecture– William Stallings Pearson Education , Eighth Edition
R2	Computer Architecture and Organization-John Hayes- Mc GrawHill-1998.
R3	Computer System Architecture -Morris Mano- Prentice Hall of India-2002.
R4	Digital Principles and Applications – by Malvino & Leach ,McGraw-Hill,
R5	Fundamentals of digital circuits - A. Anand Kumar, PHI Learning Pvt.Ltd.,2005.
R6	Digital computer fundamentals - Thomas. C. Bartee, McGraw-Hill ,1985

Online Resources

Sl No	Website Link
1	http://nptel.ac.in/course:s/Webcourse-contents/IIT-%20Guwahati/comp_org_arc/web/
2	https://en.wikipedia.org/wiki/Von_Neumann_architecture
3	https://www.javatpoint.com/classification-of-memory
4	https://www.ecs.csun.edu/~cov/comp222s16/notes/CH14PPT.pdf